JAIKUMAR SELVARAJ

Senior HPC Administrator

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Results-driven Senior HPC System Engineer and Slurm Administrator with over 10+ years of experience architecting, managing, and optimizing large-scale compute infrastructure and batch scheduling systems. Deep functional expertise across Slurm, IBM Spectrum LSF, GPU-accelerated workloads, Lustre PFS, and high-speed InfiniBand networks. Proven track record of enhancing operational efficiency through Python and Shell automation, streamlining cluster provisioning, and driving performance tuning for TOP500 supercomputing and semiconductor design environments. Adept at translating complex technical requirements into high-availability infrastructure solutions that maximize resource utilization, reduce runtimes, and accelerate enterprise research outcomes.

Professional Summary

- 1. Administered enterprise HPC batch scheduling environments utilizing Slurm and IBM Spectrum LSF, executing partition and QOS configurations, slurmdbd accounting setup, and production job failure troubleshooting.
- 2. Supported systems administration, continuous availability, and capacity planning for a TOP500-ranked supercomputing cluster (Discovery 6).
- 3. Engineered NVIDIA GPU-accelerated HPC solutions and cluster infrastructure, reducing workload runtimes, token usage, and overall compute costs.
- 4. Optimized high-throughput storage and networking by tuning Lustre Parallel File System (PFS) stripe configurations and integrating high-speed InfiniBand interconnects.
- 5. Streamlined cluster provisioning and administrative workflows by developing automated deployment scripts using Python, Bash, Shell, Ansible, and Chef.
- 6. Conducted performance tuning and benchmarking for parallel applications leveraging MPI and OpenMP to optimize communication patterns, ranks, and threads.
- 7. Delivered 24/7 L3 Linux infrastructure support and proactive cluster health monitoring using Nagios, Ganglia, and IBM Spectrum RTM to maximize system uptime.
- 8. Integrated specialized HPC technologies into semiconductor design and EDA workflows through direct collaboration with engineering groups and vendor partners.

Work Experience

HPC System Engineer Jun 2024 - Feb 2026

Exxonmobil Service & Technology (Bangalore)

Responsible for managing, automating, and tuning large-scale HPC infrastructure, executing capacity planning, standardizing node provisioning through Kickstart automation, and evaluating emerging compute technologies to support enterprise research workloads.

- Optimized high-performance computing workloads through targeted performance tuning, enhancing computational throughput and workload execution efficiency.
- Automated node provisioning and cluster OS image deployments using Kickstart and custom scripts, reducing manual configuration efforts and node integration timelines.
- Developed robust Bash, Python, and Shell automation workflows to streamline routine system maintenance tasks and cluster management procedures.
- Executed detailed capacity planning and resource allocation analyses to ensure infrastructure scalability and satisfy expanding compute demands.
- Evaluated and integrated emerging HPC technologies into production environments, modernizing cluster infrastructure while lowering total cost of ownership.Designed and configured AWS Virtual Private Cloud (VPC) infrastructure, including public/private subnets, route tables, Internet Gateways, NAT Gateways, security groups and network connectivity.
- Provisioned and managed AWS infrastructure using Terraform (Infrastructure as Code), ensuring consistent, repeatable and automated deployments.
- Designed and implemented CI/CD pipelines using AWS CodePipeline, CodeDeploy and Amazon S3 to automate application build, deployment and release workflows.
- Configured and managed CodeDeploy agents on Amazon Linux EC2 instances to automate application deployment and updates.
- Implemented infrastructure and application monitoring using Prometheus and Grafana to monitor server health, CPU, memory, disk utilization and application performance.
- Configured Grafana dashboards and Prometheus metrics collection to provide infrastructure visibility, performance analysis and proactive issue identification.

Achievements:

- Integrated novel HPC technologies that significantly elevated system compute capabilities while lowering infrastructure costs.

Technical Skills

Slurm Administration	●●●●◐
Linux Administration	●●●●◐
Shell Scripting	●●●●◐
Python Scripting	●●●●◐
Lustre PFS	●●●●◐
GPU Computing	●●●●◐
LSF Scheduler	●●●●◐
MPI Tuning	●●●●◐
Cluster Provisioning	●●●●◐
InfiniBand Interconnects	●●●●○

Soft Skills

Problem Solving
Cross-Functional Collaboration
Technical Mentorship
Stakeholder Management

Core Competencies

Batch Job Scheduling
HPC Architecture & Design
Performance Optimization & Tuning
Infrastructure Automation
Capacity Planning & Scaling
Workload Orchestration
Parallel File System Management
High Availability & Disaster Recovery
Vendor & OEM Management
Incident & Escalation Management

Certifications

- **RHCE (Red Hat Certified Engineer)**
  Red Hat
- **NVIDIA Tesla Certified Solution Architect**
  NVIDIA
- **CKA (Certified Kubernetes Administrator)**
- **Microsoft Certified: Azure Fundamentals**
  Microsoft

- Generated standardized system documentation and architecture guidelines, ensuring operational compliance and accelerating knowledge transfer across engineering teams.

Senior Staff System Information Engineer Jan 2023 - Sep 2023

Synaptics Engineering Pvt Ltd (Bangalore)

Responsible for aligning HPC capabilities with semiconductor design requirements, maintaining Linux infrastructure reliability, collaborating with vendors and OEMs, and supporting engineering teams to maximize workflow performance.

- Synchronized HPC cluster capabilities with semiconductor design requirements by collaborating closely with internal engineering, IT, and research teams.
- Partnered with hardware vendors and OEMs to integrate modern compute solutions into semiconductor design workflows, improving operational efficiency.
- Maintained continuous L3 support for Linux infrastructure and cluster environments, ensuring high system availability and minimal operational downtime.
- Conducted regular technical reviews with engineering support teams to proactively identify, diagnose, and resolve infrastructure bottlenecks.

Achievements:

- Integrated modern HPC solutions across semiconductor design processes, resulting in optimized workflow performance and enhanced engineering efficiency.
- Improved overall Linux infrastructure and cluster environment reliability through proactive maintenance and rapid issue resolution.

Senior HPC Engineer Jul 2022 - Jan 2023

Infobell IT

Responsible for executing hardware benchmark evaluations with AMD, training and mentoring junior HPC engineers on cluster architecture and scheduling systems, and facilitating technical knowledge transfer.

- Collaborated directly with AMD to execute hardware and workload benchmarks, ensuring precise performance evaluations and hardware validation.
- Mentored junior HPC engineers on cluster architecture, batch job scheduling techniques, and system diagnostics methodologies.
- Organized structured technical walkthroughs and knowledge-sharing sessions to elevate team technical capabilities and operational efficiency.
- Standardized diagnostic procedures and architecture guidelines to improve system troubleshooting speed across compute environments.

Achievements:

- Delivered accurate performance benchmarks in partnership with AMD to inform compute infrastructure optimization.
- Enhanced overall engineering team competency in cluster architecture, job scheduling, and system diagnostics through technical training initiatives.

Senior Professional at Emerging Practices Sep 2020 - Jun 2022

DXC Technology

Responsible for delivering L3 VPC Linux support, automating server deployments, managing VMware vCenter virtualization, configuring disaster recovery systems, and maintaining technical configuration data integrity.

- Delivered L3 Linux infrastructure support across Virtual Private Cloud (VPC) environments, maintaining system availability and operational health.
- Automated Linux build and deployment processes using custom Shell scripts and HPSA automation tools, drastically accelerating multi-node provisioning.
- Administered VMware vCenter operations, managing storage allocation, network VLAN configurations, and virtual machine instances to support infrastructure scalability.
- Collaborated with networking teams to diagnose and resolve complex network connectivity and performance bottlenecks.
- Configured robust system backup and disaster recovery solutions to safeguard critical enterprise data and minimize potential operational loss.
- Maintained CMDB data integrity by implementing bulk load procedures for server asset tracking and validating configuration compliance.

Achievements:

- Accelerated server deployment timelines by automating Linux OS installation procedures through custom scripting and HPSA toolsets.
- Strengthened system disaster recovery capabilities through the successful configuration and deployment of enterprise backup solutions.
- Elevated team operational expertise by developing and conducting frequent technical training programs on emerging practices.

Senior System Engineer Feb 2019 - Jan 2020

Merck KGaA

Responsible for Slurm batch scheduler administration, CPU/GPU cluster solution design, Lustre parallel file system tuning, MPI/OpenMP application optimization, and maintaining cluster availability.

- Administered Slurm Job Scheduler across heterogeneous CPU and GPU clusters, configuring job priority queues, resource limits, and workload management policies.
- Architected and deployed NVIDIA GPU-accelerated HPC infrastructure, tailoring hardware and software configurations to meet demanding scientific research requirements.
- Managed Lustre Parallel File System (PFS) storage, tuning stripe size and stripe count parameters to resolve I/O bottlenecks and optimize application throughput.
- Integrated parallel computing applications leveraging MPI and OpenMP, performing performance bottleneck analyses to accelerate compute execution.
- Maintained peak cluster availability by implementing proactive system health monitoring, automated routine checkups, and disaster recovery backup procedures.
- Translated business and research requirements into actionable technical project plans, integrating high-level scientific applications with web portal interfaces.

Achievements:

- Delivered optimized GPU solutions for HPC applications, significantly reducing simulation runtimes, token usage, and operational infrastructure costs.
- Maximized cluster availability and minimized unplanned downtime through structured weekly and monthly system health audits.
- Enhanced enterprise workflow efficiency by translating scientific research needs into targeted HPC infrastructure initiatives that supported revenue growth.

Senior System Engineer Jul 2016 - Feb 2019

Altair Engineering India Pvt Ltd.

Responsible for Altair PBSPro workload scheduler administration, PBS Works solution customization, Python/Chef automation development, and managing AWS, Docker, and hypervisor environments.

- Administered Altair PBSPro workload manager and Compute/Display Manager, configuring job scheduling parameters and resource allocation strategies.
- Developed automated workflows and application definitions using Python APIs and Bash scripts to streamline routine operational tasks.
- Customized and configured PBS Works software solutions to satisfy specific customer workload requirements and integrate with enterprise systems.
- Managed hybrid cloud and virtualized infrastructure across AWS, Docker container environments, Citrix XenServer, and KVM hypervisors.
- Leveraged PBS Analytics tools to evaluate cluster usage patterns, generate system optimization reports, and execute HPC application benchmarks.
- Implemented automation strategies using Chef configuration management to automate server setups and reduce manual configuration errors.

Achievements:

- Delivered customized PBS Works deployments that achieved seamless integration with existing client infrastructure.
- Decreased manual administrative overhead by developing Python API automation scripts and leveraging Chef configuration management.

Technical Service Specialist - GTS Dec 2014 - Jul 2016

IBM India Pvt Ltd. (Bangalore)

Responsible for supporting IBM Spectrum LSF job scheduler and PAC portal operations, managing application licensing, creating custom Shell monitoring scripts, and resolving cluster incident tickets.

- Supported IBM Spectrum LSF job scheduler operations and Platform Application Center (PAC) job submission portal, ensuring efficient job scheduling and resource management.
- Engineered custom Shell scripts for application submission templates, reducing manual user intervention during workload execution.
- Monitored cluster stability, disk utilization, and node availability using IBM Spectrum RTM and automated notification scripts.
- Executed software installations, license manager configurations, and performance benchmarking for enterprise HPC applications.
- Managed technical support tickets using Kayako Fusion, documenting standard configurations and troubleshooting procedures in internal knowledge base repositories.

Achievements:

- Streamlined technical problem-solving and cross-team communication through regular engagement in client, onshore, and offshore operations meetings.
- Expanded cluster administration capabilities into big data technologies through formal training in Hadoop ecosystem administration.

Engineer - Research & Development Jan 2011 - Dec 2014

HCL Info Systems Pvt. Ltd (Chennai)

Responsible for designing HPC system architecture, configuring High Availability server clusters, implementing Lustre PFS and InfiniBand interconnects, compiling parallel computing environments, and delivering onsite technical installations.

- Architected and deployed client-tailored HPC infrastructure, configuring high availability (HA) server clusters to ensure continuous operational uptime.
- Implemented Lustre Parallel File System (PFS) across storage nodes with storage quota policies to optimize computational data throughput.
- Integrated InfiniBand high-speed fabric interconnects and parallel runtime environments (MPI, MVAPICH, OpenMP) using Intel compilers.
- Automated software compilation and scientific package distribution across multi-node clusters using modular Shell scripts.
- Monitored cluster health metrics using Ganglia and Nagios while performing onsite hardware installations and client technical training.

Achievements:

- Improved storage system efficiency and data processing speeds by implementing Lustre Parallel File Systems across compute clusters.
- Facilitated ultra-fast inter-node data transfer rates by integrating InfiniBand network fabrics into production HPC systems.
- Enhanced system automation by developing custom Shell scripts for routine cluster maintenance and automated package deployments.

Jr Server Admin Nov 2009 - Dec 2010

Square Brothers Information Technology Pvt Ltd (Chennai)

Responsible for CentOS server administration, LVM storage management, Nagios monitoring, NFS/NIS network configuration, and server security hardening across Virtual Private Servers.

- Administered CentOS Linux distributions on Virtual Private Servers (VPS), managing LVM storage configurations including volume group expansions and shrinks.
- Configured and integrated Nagios monitoring systems to track remote host performance, service health, and automated system alerts.
- Implemented network storage and directory services including NFS, NIS, and Samba according to specific client infrastructure requirements.
- Enforced server hardening protocols, scheduled automated cron maintenance tasks, and managed system log rotations to secure server operations.

- Delivered 24/7 server monitoring and L2 support to diagnose web server issues and maintain continuous service uptime.

Achievements:

- Maintained high server availability and system uptime through proactive 24/7 Nagios service tracking and rapid issue resolution.
- Fortified VPS system security profiles by implementing baseline server hardening protocols and routine vulnerability monitoring.

Education

B.Tech / I.T (2005 - 2009)
SCAD Engineering College, affiliated with Anna University

HSC (2005)
St. Xavier's Hr. Sec School

Hobbies

Open Source Contributing, Home Lab Experimentation, Technical Blogging, Physical Fitness